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----- French -----

Exercise2:

- 1). Ecrire une fonction récursive *Triangle_Pascal* permettant de calculer les éléments du triangle de pascal de degré n.
- 2). Allouer dynamiquement la mémoire pour le tableau renvoyant le résultat.
- 3). Ecrire un programme principal de test.

----- English -----

Exercise2:

- 1). Write a recursive function *Triangle_Pascal* allowing to calculate the elements of Pascal's triangle of degree n.
- 2). Dynamically allocate memory for array returning the result.
- 3). Write a main test program.

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#include <stdio.h>
#include <stdlib.h>

int **allocate(int **pascal, int rows);

int get_pascal(int **pascal, int i, int j);

int main()
{
    int i, j, **pascal, rows, result, space;
    printf("Rows = ");
    scanf("%d", &rows);
    pascal = allocate(pascal, rows);
    for (i = 0; i <= rows; i++)
    {
        for (space = 0; space < rows - i; space++)
            printf(" ");
        for (j = 0; j <= i; j++)
        {
            result = get_pascal(pascal, i, j);
            printf("%4d", result);
        }
        printf("\n");
    }
    return 0;
}

int **allocate(int **pascal, int rows)
{
    int i;
    pascal = (int **)malloc(rows * sizeof(int *));
    for (i = 0; i < rows; i++)
        pascal[i] = (int *)malloc(rows * sizeof(int));
    return pascal;
}

int get_pascal(int **pascal, int i, int j)
{
    int space;

    if (i == j || j == 0)

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```
    return 1;
else
    return get_pascal(pascal, i - 1, j - 1) + get_pascal(pascal, i - 1, j);
}
```

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